Phase 5: Healthcare Business Questions & Visualisation Report

Analytical Inquiries & Chart Specification Mapping

1. Core Healthcare Business Questions

• BQ-1: How does chronic condition prevalence scale across adult age brackets?

• BQ-2: What proportion of patients exhibit multi-morbidity (heart disease + diabetes + stroke)?

• BQ-3: How do systolic blood pressure and fasting glucose distribute across clinical risk tiers?

• BQ-4: What is the combined cardiovascular impact of active smoking and sedentary habits?

2. Visualisation Mapping

Business Question	Chart Type Selected	Target Variables	Primary Clinical Finding
BQ-1: Age vs Disease	Clustered Bar Chart	Age Group vs HD, DM, Stroke %	Prevalence surges past age 45, reaching 89-100% at 61-80
BQ-2: Multi-morbidity	Burden Histogram	Chronic Condition Count (0-3)	1,207 patients (24.1%) suffer from the complete disease triad
BQ-3: Biomarker Clusters	Stratified Scatter Plot	Blood Pressure vs Glucose by Tier	Critical tier patients cluster above 140 mmHg and 126 mg/dL
BQ-4: Lifestyle Risk	Cross-Tabulated Bar	Smoking & Exercise Level vs HD	Sedentary smokers experience 18% higher cardiovascular incidence